

DECODED

EXPLAINED

29th June, 2026

SHARING WATERS

Sharing waters

The Tungabhadra dam is a good example of inter-State cooperation

The coming together of the Chief Ministers of Karnataka (D.K. Shivakumar), Telangana (A. Revanth Reddy) and Andhra Pradesh (N. Chandrababu Naidu), along with Union Jal Shakti Minister C.R. Patil, at the inauguration of the 33 spillway gates of the **Tungabhadra dam** on June 25 was significant beyond the optics. The three leaders pledged greater inter-State cooperation, which is welcome given that the dam has largely remained free of major disputes due to an established water-sharing formula and the Tungabhadra Board's regulation of releases. Located in Karnataka's Koppal district, the dam is considered to be the lifeline of the three southern States, providing irrigation to about 16.4 lakh acres — 9.26 lakh in Karnataka, 6.25 lakh in Andhra Pradesh, and 87,000 in Telangana. It hit the headlines in August 2024 when a crest gate was washed away during heavy inflows — which also happened when the dam was full with 105 thousand million cubic feet (tmc ft) of water. The damaged gate was replaced with a temporary one, but with much water wasted, the authorities, who did not want to take chances with the other 32 gates, chose to install high-grade steel gates, at ₹51 crore and which are expected to last 60 years.

Notwithstanding the bonhomie, the differences among the leaders on some issues concerning the larger Tungabhadra project cannot be brushed aside. The **Upper Bhadra project**, a major lift irrigation scheme under implementation in the central region of Karnataka and upstream of the Tungabhadra dam, has emerged as an irritant for Andhra Pradesh and Telangana. At one stage, the BJP-led Union government had made a provision of ₹5,300 crore in its budget for 2023-24, a few months before the Karnataka Assembly elections. However, the Centre later chose not to include it under any of its schemes, but this is being implemented by a Karnataka government undertaking. It remains to be seen whether the spirit of camaraderie will help the three neighbours to overcome their differences on the Upper Bhadra issue. The Tungabhadra dam is also facing excessive siltation, which has reduced its storage capacity from the original 133 tmc ft to about 106 tmc ft. Mr. Patil has assured the Chief Ministers about the Centre's plan to remove silt from water reservoirs across the country including the Tungabhadra. More importantly, the Union government should closely monitor and expedite, wherever required, the execution of dam rehabilitation and improvement projects being taken up in 19 States. Any laxity with regard to dam safety may have disastrous consequences. The focus must be on prevention, not reparation.



SHARING WATERS

- On **25 June 2026**, the Chief Ministers of **Karnataka, Andhra Pradesh and Telangana**, along with the Union Jal Shakti Minister, jointly inaugurated **33 new spillway gates** of the Tungabhadra Dam.
 - The event symbolised **cooperative federalism** despite differences over river water issues.
-
- **Tungabhadra river**
 - a. Formed by the confluence of Tunga and Bhadra river
 - b. Origin western ghats Karnataka
 - c. Flows through - Karnataka and Andhra Pradesh and finally joins Krishna river
 - d. Major tributaries- Vedavati, Hagari, Kumudvati, Varada
 - e. Tungabhadra dam Koppal/Vijayanagara region, states benefited Karnataka , Andhra and Telangana

SHARING WATERS

Why was it in News?

In August 2024

- Crest Gate No.19 was washed away during heavy floods.
- Large quantity of water was lost.
- Safety concerns were raised.

Government response

- Temporary gate installed.
- Technical inspection.
- Replacement of all **33 spillway gates** costing around **₹51 crore**.
- New gates expected to last about **60 years**.

SHARING WATER

Tungabhadra Board

- Established by Government of India after the project.
- Article 262 Parliament may provide for settlement of interstate river disputes.
- Inter state river and river valleys in **Union list**
- Water is a **State Subject**
- Union can regulate interstate rivers in public interest.

Important Acts

- Inter-State River Water Disputes Act, 1956
- River Boards Act, 1956
- Dam Safety Act, 2021

SHARING WATER

RIVERS AND STATES DISPUTED

RIVER	STATES
Cauvery	Karnataka,Tamil Nadu
Krishna	Karnataka, Telangana, Andra
Ravi Beas	Punjab. Haryana
Mahanadi	Odisha, Chhattisgarh

PYQ

Q. With reference to the Tungabhadra River, consider the following statements:

- 1.It is formed by the confluence of the Tunga and Bhadra rivers, both originating in the Western Ghats.
- 2.The river flows through the states of Karnataka and Andhra Pradesh before joining the Krishna River.
- 3.The Tungabhadra River ultimately drains into the Bay of Bengal independently without joining any major river.

Which of the statements given above is/are correct?

[A] 1 and 2 only

[B] 2 and 3 only

[C] 1 and 3 only

[D] 1, 2 and 3

Life on Mars: enduring mystery

Vasudevan Mukunth

In a new study in *Science Advances*, an international team of researchers has reported finding organic matter on Mars. Using the NASA Perseverance rover, the team recorded a distribution of complex organic carbon within an ancient river valley on the red planet's Jezero Crater.

Using the rover's SHERLOC instrument, the researchers detected a complex and hardy form of organic matter called macromolecular carbon within fine-grained mudstones. According to the study paper, published in *Science Advances*, this is significant as it represents the most robust detection of organic material in Jezero Crater to date and is the first time such a material has been found directly on a natural rock surface on Mars.

Over the last decade, the NASA Curiosity rover has found organic molecules in Gale Crater, proving the building blocks of life could be preserved in Mars's lakebed rocks. However, Perseverance's previous scans of the floor of Jezero Crater only showed localised and faint hints of organic materials.

The new evidence, from the Naretva Vallis channel, suggests organic matter could be more widespread on Mars than thought. The fact that these organic materials have also been found alongside minerals like carbonates and



A selfie taken by NASA's Perseverance Mars rover in 2025 shows the Cheyava Falls rock to the rover's left. NASA

sulphates suggests they may have been trapped and preserved by water-driven processes billions of years ago. While the new study noted that these organics could have been formed by geological (i.e. abiotic) processes rather than biological ones, the findings are, in the paper's words, "astrobiologically compelling", showing that complex carbon can survive the planet's harsh radiation for aeons.



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LIFE ON MARS

Why in News?

Recent Discovery

- A study published in the journal **Science Advances** reported the detection of **complex organic matter** on Mars.
- The discovery was made by NASA's **Perseverance Rover** in **Jezero Crater**.
- It is the strongest evidence so far of preserved organic material on the Martian surface.

Why is this important?

- It does **NOT** prove life existed on Mars.
- It suggests Mars once had conditions that may have supported life.

LIFE ON MARS

Basic Facts

- Fourth planet from the Sun
- Called the **Red Planet**
- Red colour due to **iron oxide (rust)**
- Two moons:
 - Phobos
 - Deimos

Interesting Facts

- One day = 24.6 hours
- One year = 687 Earth days
- Thin atmosphere made mostly of carbon dioxide (CO₂)

Why is Mars Important?

Scientists believe that:

- Mars once had rivers and lakes.
- It may have had a thicker atmosphere.
- Water existed billions of years ago.

Therefore scientists ask:

Could life have existed on Mars?

This is the field of **Astrobiology**

Found complex organic compounds

LIFE ON MARS

What is Jezero Crater?

Scientists believe

- Jezero Crater was once a lake.
- Rivers flowed into it billions of years ago.
- Sediments settled there.

Therefore

It is one of the best places to search for ancient life.

LIFE ON MARS

Geological vs Biological Processes

- Geological process- abiotic process

Without life- volcanoes, chemical reactions, meteorite impacts

- Biological process

Produced by living organisms

- Minerals found—

carbonates - usually form in water

Sulphates- can preserve ancient organic material

These minerals suggest that liquid water once existed

LIFE ON MARS

ISRO Mars Orbiter Mission (Mangalyaan)

Launch-2013

Reached Mars 2014

Achievements

- First Asian nation to reach Mars orbit.
- First country to succeed on its first attempt.
- Low-cost successful mission.

Current status –Mission concluded after fuel depletion and communication loss in 2022.

Interstellar movie and Mangalyan

LIFE ON MARS

Missions to Mars

VIKING	USA
CURIOSITY	USA
PERSEVERANCE	USA
HOPE	UAE
Tianwen 1	China
Mangalayan	India

PYQ

Consider the following statements:

The Mangalyaan launched by ISRO

- 1. is also called the Mars Orbiter Mission**
- 2. made India the second country to have a spacecraft orbit the Mars after the USA**
- 3. made India the only country to be successful in making its spacecraft orbit the Mars in its very first attempt**

Which of the statements given above is/are correct?

1. 1 only

2. 2 and 3 only

3. 1 and 3 only

4. 1, 2 and 3

Buyer beware

Tracking down defective batches of drugs is essential

The Health Ministry's notification to expand Schedule H2 drugs to entire therapeutic classes now from a curated list of brands earlier is a commendable change from regulating based on revenue to regulating based on risk. The government had introduced Schedule H2 in 2022-23, which required a barcode or QR code on each pack of specific drugs to verify the pack's authenticity. It is part of a framework that also touches on regulating active pharmaceutical ingredients and export-oriented pharmaceutical compliance. Now, with the additional symbols encoding a product identifier, the manufacturing licence number, batch number, and other details, authorities hope to better track down defective batches. Counterfeit networks have long targeted vaccines, cancer medicines and antimicrobials, with WHO flagging high volumes of fake antimicrobials in low- and middle-income countries. India has one of the world's highest antimicrobial resistance rates, and sub-standard antimicrobials can lead to sub-therapeutic dosing, exerting selection pressure on resistant strains. The Narcotics Control Bureau has also expressed concerns about medicinal opioids and psychotropic substances 'leaking' into illicit markets. The government will hope that the new system addresses the U.S. FDA's and the European Medicines Agency's recurring concerns about quality control, an issue on which India has found itself on the back foot over contaminated cough syrups. The U.S. Trade Representative has also repeatedly identified India as a leading source of counterfeit medicines, with many fake pharmaceuticals seized at the U.S. border originating from or transiting through India.

Together with policy changes effected by the Jan Vishwas Act 2026, the new framework finally distinguishes between procedural non-compliance and substantial non-compliance, considering only the latter constitutes meaningful enforcement. In the long term, the government should rationalise the compliance burden and eliminate the risk of corruption associated with discretionary enforcement. The government must also recognise that the QR code system will only be effective if backed by a state-managed database that pharmacists and regulators can access in real time, supported by interoperable software and scanning infrastructure across States. Second, pharmacists and consumers must develop a habit of verifying medicines before a sale. Third, the compliance needs imposed by new packaging requirements and the IT integration will strain MSME manufacturers in particular. Finally, prescription data on controlled substances is sensitive information and will need to be handled with a digital governance layer that does not yet exist. Thus, whether the new framework will improve India's reputation as a pharmacy comes down to its implementation.

BUYER BEWARE

SCHEDULE H2 MEDICINES

Schedule H2 is a government regulation requiring certain medicines to carry a **unique QR code or barcode**.

The code contains:

- Product ID
- Manufacturer name
- Manufacturing licence number
- Batch number
- Expiry date
- Authentication details

Purpose

Every medicine can be traced from factory to patient.

BUYER BEWARE

QR CODE

Manufacturer generates QR code -> Medicine at distributor -> pharmacy scan QR code -> consumer verifies it using smart phone

Drug traceability

factory-> warehouse-> distributor -> pharmacy-> patient

API active pharmaceutical ingredient

What are Fake pharmaceutical drugs , implications

RELATED GOVT INITIATIVES

- **Central Drugs Standard Control Organisation (CDSCO)** – India's national drug regulator.
- **Pharmacovigilance Programme of India (PvPI)** – Monitors adverse drug reactions.
- **Ayushman Bharat** – Expands access to healthcare. e- sanjeevani
- **Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP)** – Provides affordable generic medicines.
- **National Digital Health Mission (Ayushman Bharat Digital Mission)** – Promotes digital health records and interoperable health systems.

The new digital slavery needs constitutional guardrails

In his profound encyclical, *Magnifica Humanitas*, Pope Leo XIV grounds the contemporary crisis of technology in elemental theology, reminding humanity that the individual stands at the absolute centre of divine creation. Yet, this sacred human dignity faces an unprecedented challenge. The Holy Father warns that artificial intelligence (AI), left unchecked, threatens to turn the ownership of our personal data into a new form of digital slavery – a subjugation as cruel as human bondage. Because the stakes are so inherently tied to human dignity, the Pontiff insists that the governance of AI cannot rely on vague good intentions, nor can it be left to the private consciences of the engineers who build these systems. Instead, he issues an urgent call for robust, binding law over abstract ethics, independent public oversight over the empty promises of Silicon Valley's modern pharaohs – and a clear requirement that a human being remains accountable whenever an automated system decides who gets a loan, a job, a medical bed, or an education. Pope Leo XIV's core demand is that this new electronic Curia be strictly regulated rather than quietly surrendered to a handful of private tech monopolies whose reach and resources already outstrip most sovereign governments. He even goes so far as to commend the courage to deliberately slow down the development of AI.

When regulation lags

However, a fundamental problem remains: AI and morality do not observe the same time. AI is developed at the breakneck speed of the start-up culture, driven by an ethos to move fast and run perpetual beta tests on society in the guise of progress. This velocity is fuelled not just by commercial rivalries, but by relentless mathematical innovation spanning from Silicon Valley to Shenzhen. It presents a deeper hurdle for lawmakers worldwide: parliament can govern what a person does, but it can never forbid a mathematical theorem, a discovery, or an equation from being made. How do lawmakers fulfil the Pope's wishes?

Law in most democracies moves slowly. By the time landmark legislation such as the European Union's Artificial Intelligence Act or the United Kingdom's Online Safety Act were painstakingly debated and passed, the specific technical harms they were built to combat had already mutated, leaving an entire generation to grow up inside unaddressed digital vulnerabilities. Policymakers have historically been behind the innovation curve, and under current paradigms, we will be again.

If legislation is fated to lag behind innovation, the consequences for democratic societies are not merely regulatory; they are existential. Democracy cannot function if citizens cannot distinguish reality from fabrication. At its core, democratic governance relies on a shared epistemic foundation: a collective agreement on



Shashi Tharoor

Fourth-term Member of Parliament (Lok Sabha) for Thiruvananthapuram (Congress party), the Chairman of the Parliamentary Standing Committee on External Affairs and the Sahitya Akademi Award-winning author of 29 books, including, most recently, *The Sage Who Reimagined Hinduism: The Life, Lessons and Legacy of Sree Narayana Guru*

Unchecked AI threatens democracy, demanding constitutional safeguards beyond conventional regulation

basic facts from which public debate, policy, and electoral choices can flow. When that foundation is systematically undermined, the entire democratic project is threatened. Today, AI-generated disinformation and advanced synthetic media ("deepfakes") have advanced to a level of fidelity where the human eye and ear can no longer reliably detect forgery. This is already our reality. Highly convincing audio and video duplications of political leaders are deployed strategically during sensitive electoral cycles to depress voter turnout, fabricate scandals out of thin air, and instantly shatter public trust in legitimate state institutions.

This vulnerability is hyper-charged by algorithmic manipulation. Big Tech platforms operate on business models engineered exclusively to maximise user engagement. Because outrage, fear, and sensationalism generate the highest click-through rates, platform algorithms systematically prioritise and amplify hyper-partisan content, driving radicalisation. By trapping citizens within hyper-customised echo chambers, these systems normalise online hate and accelerate social fragmentation. In doing so, private platforms exert an unprecedented form of unaccountable power, effectively rewriting the rules of the public square to optimise quarterly corporate profits at the expense of social cohesion.

Democracy's digital vulnerability

When a society is deeply polarised and stripped of a shared reality, it becomes a soft target for foreign interference. Adversarial nation-states and non-state actors have weaponised these platform vulnerabilities, transforming them into a theatre for sophisticated information warfare. Foreign information manipulation operations are no longer clumsy, bot-driven spam campaigns; they are highly targeted, AI-driven psychological operations designed to covertly exploit pre-existing religious, ethnic, or socioeconomic fault lines within a target nation. By covertly funding, generating, and amplifying divisive narratives, hostile foreign actors can destabilise a democracy from within, turning its own citizens against one another. These coordinated campaigns represent nothing less than a direct, strategic threat to democratic sovereignty itself. As the world's largest democracy and a global technology hub, India stands at the absolute epicentre of this crisis. In an environment where digital adoption outpaces structural digital literacy, the weaponisation of synthetic media and algorithmic polarisation poses a unique threat to India's pluralistic society.

If traditional, reactive legislation is doomed to always trail behind mathematical innovation, India cannot rely on standard, backward-looking regulatory frameworks. Instead, an enduring law and policy framework for India must be anchored in five foundational pillars that operate concurrently.

First, AI governance must adopt a strictly

rights-based framework that prioritises individual human dignity and digital autonomy, ensuring citizens possess unalienable rights over their personal data, strict consent protocols, and clear protections against algorithmic discrimination in critical sectors such as employment, credit, and health care. Second, platforms must be subjected to genuine democratic accountability; large technology firms can no longer hide behind absolute safe-harbour immunities while their algorithms profit from the viral spread of destabilising disinformation. They must be legally compelled to introduce structural transparency, allowing independent audits of their recommendation engines and forcing them to accept systemic liability for algorithmic amplification that results in real-world violence.

Third, even as these frameworks are constructed, free speech protections must remain fully intact. The power to define and combat disinformation must never devolve into a tool for state-sponsored censorship or the silencing of political dissent. The focus of regulation must remain strictly on structural platform mechanics, such as automated bot networks and deepfake originators, rather than the heavy-handed policing of individual ideological speech.

Fourth, because technical fixes are entirely insufficient without building cognitive resilience within the populace, India must launch a massive, state-backed educational initiative focused on media literacy and digital citizenship. This curriculum must be integrated across schools, universities, and rural community centres to train citizens to critically evaluate digital sources and identify emotional manipulation tactics. Finally, to defend national sovereignty, India must establish sophisticated, cross-sector early-warning systems capable of detecting coordinated misinformation operations in real-time. By leveraging advanced detection tools and fostering deep collaboration between state security apparatuses, independent fact-checking networks, and ethical hackers, the nation can identify and neutralise foreign information warfare campaigns before they achieve viral velocity.

A constitutional imperative

Ultimately, the lesson of our era is clear: AI governance cannot remain merely regulatory or technical. It is a profound mistake to treat the manipulation of the information ecosystem as a series of isolated technical glitches to be patched with minor corporate updates or narrow statutory tweaks. Because these technologies possess the unique capability to distort truth, polarise societies, and erode the sovereign choice of voters, AI governance must rise to the level of a constitutional imperative. The right to an unmanipulated information ecosystem, where reality can be clearly distinguished from corporate or state fabrication, must be viewed as an indispensable extension of the fundamental right to life, liberty, and free expression.

DIGITAL SLAVERY

Why in News?

Background

An editorial discusses the growing risks posed by **Artificial Intelligence (AI)** to democracy, privacy, and individual rights.

It argues that AI should not be regulated only as a technological issue but also as a **constitutional issue**.

Main Concern

AI should serve humanity—not control it.

DIGITAL SLAVERY

Artificial Intelligence (AI)

Machines performing tasks requiring human intelligence.

Machine Learning (ML)

A branch of AI where computers learn patterns from data without being explicitly programmed.

Generative AI

AI that creates new content such as text, images, videos, or music.

Algorithmic Bias

When AI produces unfair or discriminatory outcomes because of biased training data or design

DIGITAL SLAVERY

Digital Autonomy -A person's ability to make independent choices online without manipulation or undue influence.

AI Governance- The laws, policies, institutions, and ethical principles used to regulate AI safely and responsibly.

Internal Security Link

Cyber crime
Cyber terrorism
Finance scams
Identity theft
Election interfere

Challenges in AI regulation

Tech change
No global std
Cross border data flow
Protect privacy

LANDMARK JUDGEMENTS

Justice K.S. Puttaswamy v. Union of India (2017)

- Declared **Right to Privacy** a Fundamental Right.
- Very important for AI and data protection.

Anuradha Bhasin Case (2020)

- Internet access is essential for freedom of speech.

Shreya Singhal Case (2015)

- Struck down Section 66A of the IT Act.
- Protects online freedom of expression.

PUCL vs Union of India (Telephone Tapping)

- Surveillance should have legal safeguards.

Iran oil is back, but Indian refiners aren't rushing in



SUKALP SHARMA

A US waiver, issued June 22, has allowed the production, delivery and sale of Iranian oil, petroleum products and petrochemicals till August 21. The US has also committed to lifting its naval blockade of Iranian ports. Tehran is counting on both measures to move its oil — which has for years been predominantly exported to China — to the wider international market.

The sanctions waiver presents an opportunity for India, once a major buyer of Iranian crude. The National Iranian Oil Company has started reaching out to international oil companies, including Indian refiners, to resume commercial ties.

Sources in India's refining sector said the companies are in touch with the Iranian side. They are also looking into the commercial, logistical and technical feasibility of purchasing Iranian oil.

Although some barrels of Tehran's oil are bound to make their way to India over the next two months owing to discounts, industry experts and insiders don't expect large-scale purchases, at least for now.

What after the 60-day waiver

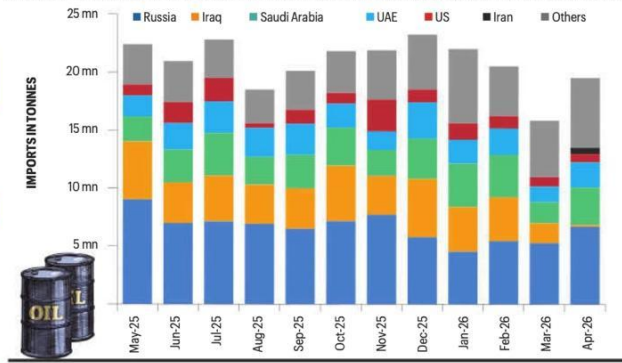
Indian refiners are expected to tread cautiously amid a fragile peace pact and lack of long-term clarity on the future of Iranian oil. There are also logistical and payment-related challenges associated with Iranian oil trade owing to US sanctions on Tehran's financial sector.

While the US waiver pertains specifically to Iran's oil, petroleum product and petrochemical exports, and not to the financial sector sanctions on Tehran, the waiver notification does mention that payments for such purchases will be allowed in dollar-denominated funds. This could help ease some, if not all, payment-related pain points. But how it would actually play out remains to be seen.

"Refiners considering Iranian crude will focus on three key factors: the durability of sanctions relief, pricing and discounts, and the availability of payment, insurance, shipping, and logistics mechanisms. Of these, payment remains the biggest

INDIA'S EVOLVING CRUDE IMPORT BASKET

The chart shows the share of different countries in crude oil imports. Iraq's share, for instance, dipped amid the West Asia war. Iran accounted for 2.7% of crude imports in April after US waived its sanctions.



SOURCE: DIRECTORATE GENERAL OF COMMERCIAL INTELLIGENCE AND STATISTICS

DATA COMPILED BY SIDDHARTH UPSANI

hurdle," said Sumit Ritolia, manager, modelling & refining, at analytics firm Kpler.

Refinery executives said that calls on Iranian oil imports will be taken once there is clarity on all these aspects. Indian refiners steer clear of oil and gas under US sanctions to avoid the risk of attracting secondary sanctions from Washington.

According to Abu Dhabi-based energy analyst Natalia Katona, any decision to restart oil trade with Iran would likely be made first in the refiners' compliance departments — not their trading desks.

"Even if crude trade is temporarily permitted, the banking side remains complicated. Iranian banks are still heavily sanctioned, so buyers would need to resolve the formalities of payment, documentation and settlement before regular trade can resume. Now that the waiver has actually been issued, Indian companies may start buying Iranian crude. But I think the number of buyers will still be quite limited," Katona told *The Indian Express*. "So far, Chinese refiners that bought sanctioned Iranian crude mostly dealt directly with Tehran and paid in yuan. That is not a model Indian refiners appear comfortable replicating."

Shaky supply

In 2009-10, Iran accounted for 14.4% of India's oil imports. As sanctions intensified, the imports tapered.

Imports briefly recovered in the mid-to-late 2010s after the Iran nuclear deal, but crashed after Trump tore it up.

April's crude oil consignment from Iran was the first in 7 years.

Signals in the previous waiver

Experts say India's buying behaviour during the month-long sanctions waiver on Iranian oil issued by the US during the West Asia war also supports the argument that it is unlikely to quickly ramp up crude imports from Iran.

After a gap of nearly seven years, Indian refiners again bought Iranian crude in April (see chart). But the volume stood at just 5,30,000 tonnes — 2.7% of India's overall oil imports for the month — according to data from the Directorate General of Commercial Intelligence and Statistics. Ritolia said the previous waiver generated limited participation from non-Chinese buyers because of payment curbs and other issues.

Further, oil scarcity was a major motivation for India during the previous waiver as it scrambled to make up for the lost supplies from other West Asian countries. But if crude from Saudi Arabia, the UAE and others starts flowing again, India has little reason to take on the compliance risk attached to Iranian barrels, said Katona.

"It is worth looking at the previous waiver, which was issued on March 20. India bought only two Iranian cargoes

under it, and they arrived in mid-April... Based on that earlier pattern, I would expect the biggest increase in Indian buying interest for the second half of July. That is probably when companies will start trying to secure as much oil as possible, while still leaving enough time to complete the payments and deliveries before the waiver expires," she said.

According to her, payments made to Iranian entities after August 21 would carry a serious compliance risk, so Indian refiners will not want to get too close to the deadline.

The China factor

China is likely to return as the dominant buyer of Iranian crude once oil flows through the Strait of Hormuz normalise. Before the crisis, China was buying almost all of Iran's oil exports. Well-oiled logistical arrangements and payment mechanisms are already in place between the Chinese buyers and the Iranian sellers. So for Indian refiners to lift Iranian oil cargoes, they would most likely have to compete with their Chinese counterparts, and perhaps even other interested buyers.

To add to that, the discounts on Iranian oil could narrow, taking away a significant price advantage from a buyer's perspective. Iran was forced to sell its oil at a discount as the US sanctions kept most potential buyers at bay, but with the waiver in place, discounts may shrink considerably — as would the scope for refiners to bargain.

"There are around five million barrels of Iranian crude already loaded on vessels in the Gulf, so Indian buyers will probably focus on cargoes that are already afloat. That would be much faster and easier operationally, the crude could reach India in about a week. At the same time, these barrels will probably be in very high demand. China will still be interested, and even US buyers may want to purchase it (under the waiver). So we could even see a situation where Indian refiners have to compete for Iranian cargoes, even though they are geographically the closest major buyers," said Katona.

It is also worth noting that India has significantly diversified its oil import basket beyond the Gulf over the past few years, and that helped the country ensure adequate supplies even during the peak of the West Asia crisis. Simply put, there is no real pressure on Indian refiners to lift Iranian oil cargoes even during a global supply squeeze.

A lot now depends on the terms Tehran offers to willing buyers.

IRAN'S OIL IS BACK

Background

- The **United States temporarily relaxed sanctions on Iranian oil** for 60 days.
- This created an opportunity for countries, including India, to resume crude oil imports from Iran.
- However, Indian refiners are proceeding cautiously due to financial, legal, and geopolitical concerns.

Why is it important?

- India imports nearly **85% of its crude oil requirement**.
- Energy security is directly linked to economic growth, inflation, and foreign policy.

India's domestic oil production is insufficient to meet demand.

India is the **third-largest consumer of crude oil** in the world.

STRAIT OF HORMUZ

Location

Between

- Iran
- Oman

Importance

- Connects the Persian Gulf to the Arabian Sea.
- Nearly **one-fifth of global oil trade** passes through this narrow waterway.



Smart AI caching can keep the data flowing when disaster strikes

In cooperative caching, different parts of a disaster-response network, including satellites, drones, base stations, and emergency vehicles, work together to store and share useful data: when one node receives or generates important content, such as satellite images or video, nearby nodes may also cache copies based on demand

Shamim Haque Mondal

Two years ago, Mundakkai and Charalmala in Kerala experienced powerful landslides that killed hundreds of people. Last year, a large village in Dhaurah, Uttarakhand, was washed away following heavy rain. In the monsoon season of 2025, the States in Northeast India were devastated by frequent floods. More than a hundred people lost their lives after heavy rain in Uttar Pradesh on May 13-14 this year.

During such disasters, it is common for telecom towers to topple over, power lines to be cut, and roads to be closed. In these situations, real-time information as to what is happening on the ground, where people are stuck, which routes are still open, etc. becomes crucial for rescue workers and medical workers. In the absence of such communication, rescue operations are delayed, more property is damaged, and more lives are lost.

Cooperative caching
'Solving' communications in disaster-affected areas is a long-standing problem. Recently, in a paper published in the *IEEE Transactions on Services Computing*, researchers from Ireland, led by Sangita Dhara of Trinity College Dublin, presented a novel approach – a way to transmit important real-time information using a technology called collaborative caching even if the local network is in a subpar condition.

At a time of disaster, the local administration typically has three main communication channels: satellite or satellite-based communications, drones or unmanned aerial vehicles, and some *ad hoc* network based on the ground. A satellite can beam data down over a wide area, so accessing that signal is relatively easy, but the big problem here is data latency, i.e. the delay in getting information. Time is of the essence. It is also possible to take pictures from the sky or transmit live video using drones. However, they are limited by their short range, limited battery capacity, and, above all, obstacles in the form of inclement weather. Finally, ground-based wireless networks can help with local communication – but they are often damaged or non-functional during disasters.

In this context, the researchers have presented cooperative caching. Here, different parts of a disaster-response network, including satellites, drones, base stations, and emergency vehicles, work together to store and share useful data. When one node receives or generates important content, such as satellite images or video, nearby nodes may also cache copies based on the demand. This cooperation allows rescue teams to



During disasters, real-time information about where people need help and which routes are still open is crucial for rescue and medical workers. REUTERS

retrieve information from the nearest available source rather than a distant origin, reducing delay, improving reliability when infrastructure is damaged, and increasing the chance that critical information is available in real time.

Automatic decision-making
Creating this caching system is technologically not simple, however. The drones are airborne, rescue vehicles are on the road, and the position of satellites is constantly changing. It is also difficult to know when and where information will be required, plus the storage of each device will be limited.

To address this problem, the researchers developed a statistical model called contextual multi-armed bandit (CMAB), an AI model that optimises caching decisions quickly.

The model learns from each of the previous decisions, then reviews three factors: what data are recently available; what data demand is currently high, and how much memory is needed in the cache. For example, it would be helpful to understand that a photo taken 10 minutes before a flood is more effective than a photo taken one hour before the flood. Similarly, video footage or a map of a place that most rescuers are asking for is more important than those that fewer ones are looking for. Sometimes, instead of caching high-resolution video – such as 4K video – short text alerts or warning messages could help rescuers learn

The researchers developed a statistical model called contextual multi-armed bandit, an AI model that optimised caching decisions quickly

productive without taking up large amounts of storage. The model is taught to make such decisions automatically.

Federated learning
While in CMAB each node learns from what information it alone has, a more advanced version called FMAB – short for federated multi-armed bandit – learns from its information as well as what the nodes nearby have learnt.

"Caching decisions are executed periodically rather than per request, amortising the computational cost," the researchers wrote in their paper. "Even the highest observed decision latency of Contextual MAB (around 87 microseconds) remains negligible compared to network delays, ensuring system complexity does not hinder the practical applicability for dynamic and real-time post-disaster deployments."

The work also highlights the importance of a three-tier network, where space, air, and ground are working together, called Space Air Ground Integrated Network (SAGIN). Here, each layer plays a unique role, but the limitations that exist between them are greatly reduced due to caching, and the system becomes more efficient. In the

post-disaster scenario, it is important not to look at all the information that is available but only at which information is more useful. An updated road map can tell us which bridges are still usable while a live video can let us know that boats are urgently needed in a specific part of the floodplain.

All together

As a result, content should be seen not only as data but as time-dependent actionable data. And the goal of advanced caching should be to learn how to store the most valuable information in the most appropriate place with limited resources. To meet this goal, SAGIN works to build external infrastructure, CMAB helps to make acceptable decisions, and FMAB can make the network stronger in the post-disaster period by spreading those decisions across multiple nodes.

However, the results of this simulation-based study depend on many parameters, such as the weather at the time, the flight of the drone, energy management, hardware malfunctions, cyber security, and the behaviour of rescuers in reality.

Moreover, it is not always technically possible to determine which content is valuable or preferable. In many cases, the needs of the local administration or humanitarian considerations can also speed up the rescue.

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THE GIST

It is common for communication to get disrupted when a disaster occurs. The absence of real-time information delays rescue operations

Researchers from Ireland found a way to transmit important real-time information using a technology called collaborative caching even if the local network is in a subpar condition

The work also highlights the importance of a three-tier network, where space, air, and ground are working together, called Space Air Ground Integrated Network (SAGIN)

AI & DISASTER MANAGEMENT

Researchers have proposed an AI-based **cooperative caching system** that helps maintain communication during natural disasters.

Why is it important?

During disasters such as:

- Floods
- Landslides
- Cyclones
- Earthquakes
- Cloudbursts

communication networks often fail, delaying rescue operations.

The new system enables **real-time information sharing** even when infrastructure is damaged.

AI & DISASTER MANAGEMENT

CACHING

Caching means **temporarily storing frequently used data** in a nearby device or server so that it can be accessed quickly.

COOPERATIVE CACHING

Instead of storing data in only one place, **multiple devices work together and share stored information.**

Devices involved:

- Satellites
- Drones
- Mobile towers
- Emergency vehicles
- Command centres

AI & DISASTER MANAGEMENT

Contextual Multi-Armed Bandit (CMAB)

What is it?

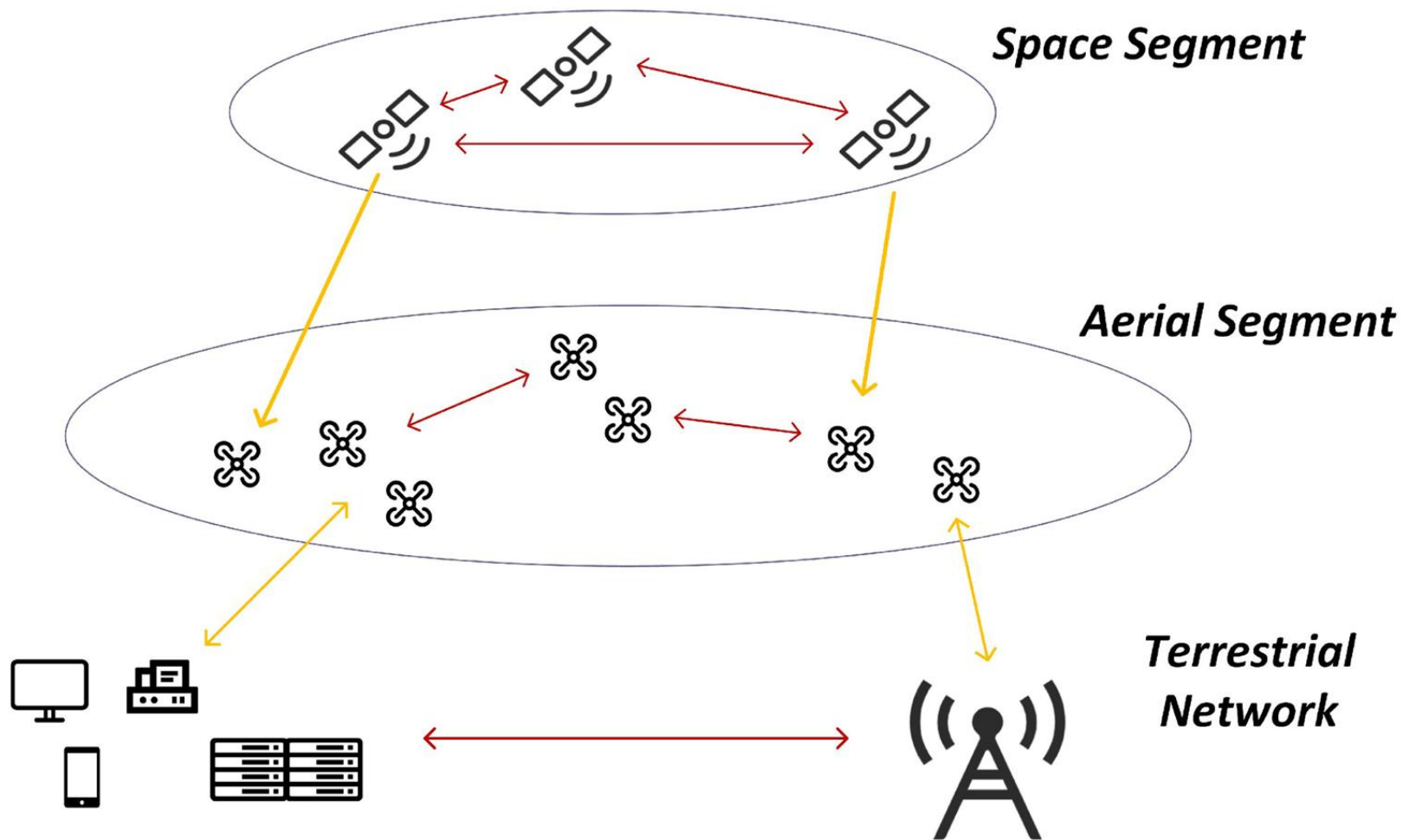
A Machine Learning algorithm that helps choose the **best decision** based on the current situation.

SAGIN (Space-Air-Ground Integrated Network)

Space - satellites

Air- drones

Ground - mobile towers



Thank You